(c) Compile-time error
(d) Undefined Output

Section B: AI / DL Algorithms

1. The number of nodes in the input layer is 10 and the hidden layer is 5. The maximum number of connections from the input layer to the hidden layer are

A. 50
B. less than 50
C. more than 50
D. It is an arbitrary value

2. Which of the following statements is true when you use 1×1 convolutions in a CNN?

A. It can help in dimensionality reduction
B. It can be used for feature pooling
C. It suffers less overfitting due to small kernel size
D. All of the above

3.Which of the following functions can be used as an activation function in the output layer if we wish to predict the probabilities of n classes (p1, p2..pk) such that sum of p over all n equals to 1?

A. Softmax
B. ReLu
C. Sigmoid
D. Tanh

4.The input image has been converted into a matrix of size 28 X 28 and a kernel/filter of size 7 X 7 with a stride of 1. What will be the size of the convoluted matrix?

A. 20x20
B. 21x21
C. 22x22
D. 25x25

5.Which of the following is NOT a common machine learning task?

A. Classification
B. Regression
C. Sorting
D. Clustering

6. In CNN, having max pooling always decrease the parameters?

A. True
B. False
C. Can be true and false
D. Can not say

7.What is the difference between classification and regression in supervised learning?

A. Classification predicts continuous values
B. Regression predicts categorical labels
C. Classification predicts labels
D. Regression uses unlabeled data

8. What is a key difference between supervised and unsupervised learning?

A. Supervised learning uses labeled data
B. Unsupervised learning requires labeled data
C. Supervised learning is faster
D. Unsupervised learning predicts labels

9.During training, you observe that the gradients for the initial layers of your deep network are becoming extremely small, causing learning to stall. Which activation function is well-known for contributing to this "vanishing gradient" problem?

A) ReLU
B) Leaky ReLU
C) Tanh
D) Sigmoid

10.An engineer trains a simple linear regression model to predict house prices but finds that it performs poorly on both the training and validation sets. The error is consistently high. Which of the following best describes the problem with the model?

A) High variance and low bias
B) Low variance and low bias
C) High variance and high bias
D) Low variance and high bias

11.Which statement best describes zero-shot learning?

A) The ability of a model to correctly classify instances from classes it has never seen during training.
B) A model that requires zero training data to perform any task.
C) Training a model with only a single example for each class.
D) A training technique that uses a learning rate of zero.

12. Model quantization is a technique primarily used for what purpose?

A) To reduce the model's size and improve inference speed, often by using lower-precision numbers like 8-bit integers.
B) To increase the model's accuracy on the validation set.
C) To automatically discover the optimal number of layers for a network.
D) To make the model more resistant to adversarial attacks.

13. What does the 'vanishing gradient problem' primarily affect?

A) The stability of shallow networks
B) The training of very deep networks.
C) The speed of a forward pass
D) The choice of the loss function

14. What is the primary role of an 'attention mechanism' in a deep learning model?

A) To randomly drop connections during training to prevent overfitting.
B) To allow the model to dynamically focus on the most relevant parts of the input sequence when producing an output.
C) To ensure that the gradient has a constant variance across all layers.
D) To normalize the activations within a layer to stabilize training.

15. In object detection, what does Intersection over Union (IoU) measure?

A) The model's confidence in its prediction
B) The overlap between a predicted bounding box and a ground-truth bounding box
C) The union of all objects detected in an image
D) The intersection of the training and validation datasets

16. What is the main purpose of using Non-Max Suppression (NMS) in object detection?

A) To increase the number of bounding box predictions
B) To improve the model's classification accuracy
C) To eliminate multiple, overlapping bounding boxes for the same object
D) To speed up the training process

17. What is model pruning?

A) A method for visualizing the decision boundaries of a model
B) A method for augmenting the training data to improve model generalization
C) The process of selecting the best hyperparameters for a model
D) A technique for reducing the size of a neural network by removing redundant weights or neurons

18. In a Transformer, what is the main advantage of using Multi-Head Attention instead of a single attention mechanism?

A) It significantly reduces the overall computation time
B) It allows the model to learn different types of relationships and focus on different parts of the sequence simultaneously

C) It is a regularization technique that prevents overfitting

D) It is the only way to process color images

19.In high-dimensional optimization for deep networks, why are saddle points often a more significant challenge than local minima?

A) Saddle points are more numerous and the gradient becomes very small around them, slowing down or stalling the learning process.
B) Local minima have a higher loss value than saddle points.
C) Gradient descent algorithms are mathematically incapable of escaping saddle points.
D) Saddle points only exist in convex optimization problems, which are rare in deep learning.

20. Why must positional embeddings be added to the patch embeddings in a Vision Transformer?

A. To provide spatial order, since self-attention is permutation-invariant.
B. To speed up training without affecting model understanding.
C. To guarantee better accuracy on unseen images.
D. To allow the transformer to compute attention scores.

21. How does Retrieval-Augmented Generation (RAG) enhance the capabilities of a large language model?

A. It replaces the transformer's attention layers with a retrieval mechanism.
B. It improves performance only by adding extra parameters at inference time.
C. It retrieves relevant external documents and uses them to generate more accurate, grounded answers.
D. It eliminates the need for pretraining on large text corpora.

Section -C Python Programming

1.What is the primary purpose of a decorator in Python?

a) To change the name of a function or class.
b) To add new functionality to an existing function or class without modifying it directly.
c) To create a copy of a function or class.
d) To remove a function or class from memory.

2. Which of the following statements about Python's generators is true?

a) Generators are a type of list that can be used to iterate over a sequence.
b) Generators can be reset and reused multiple times.

c) The yield keyword is used to return a value from a generator function without exiting the function.
d) Generators are created using the return keyword.

3.What is the difference between is and == in Python?

a) is compares the values of two objects, while == compares their identity.
b) is and == are the same for all immutable objects like strings and integers.
c) is checks if two variables reference the same object in memory, while == checks if their values are equal.
d) == should always be used for object comparison, as is can lead to unexpected behavior.

4. How do you access the value of a private variable __data inside a class MyClass from outside the class?

a) You cannot access a private variable from outside the class.
b) By using the syntax MyClass.__data.
c) By using the syntax _MyClass__data.
d) By using a property method to make it public..

5. What will be the output of the following code snippet?

```
a = { 'A' : 1, 'B' : 2 }

b = a.copy( )

b[ 'A' ] = 100

print( a[ 'A' ] )
```

a) 100
b) 1
c) AttributeError
d) None

6. Which of the following is an example of a list comprehension?

a) (x for x in range(10))
b) {x for x in range(10)}
c) [x for x in range(10)]
d) {x: x for x in range(10)}

7. What does the pass statement do in Python?

a) It is used to terminate a loop or conditional statement.

b) It is a placeholder for future code, doing nothing.
c) It signifies the end of a block of code.
d) It is used to raise an exception.

8.Which of the following methods is used to make an object iterable?

a) __next__()
b) __iter__()
c) __contains__()
d) __getitem__()

9. Which of the following is true about Python's GIL (Global Interpreter Lock)?

a) It allows multiple threads to execute Python bytecode in parallel on multiple CPU cores.
b) It is a feature of all Python implementations.
c) The GIL is a mutex that protects access to Python objects, preventing multiple threads from executing Python bytecode simultaneously.
d) It is a performance optimization for all I/O-bound tasks.

10.What is a lambda function in Python?

a) An anonymous function that can have multiple expressions.
b) A function that can only be defined within a class.
c) A small, anonymous function that can only contain one expression.
d) A named function used for short, one-line operations.

11. What is the correct way to raise an exception in Python?

a) Using the throw keyword followed by the exception name.
b) Using the raise keyword followed by an instance of an exception.
c) By calling the exception class's constructor.
d) Using the except keyword with a custom error message.

12. What is the purpose of the super() function in Python?

a) To access methods of a class's sibling classes in a multiple inheritance hierarchy.
b) To explicitly call the constructor of the child class.
c) To return a proxy object that delegates method calls to a parent or sibling class of a type, following the Method Resolution Order (MRO).
d) To create a new instance of a parent class.

13. What is a descriptor in Python?

a) A function that is used to validate a class's attributes.
b) A class that implements at least one of the __get__(), __set__(), or __delete__() methods.
c) An object that is used to automatically generate documentation for a class.
d) A way to make a class's attributes 'private'.

14. What is the primary purpose of the __call__() method in a Python class?.
a) To make an object callable like a function.
b) To define the class's constructor.
c) To define how the object is represented as a string.
d) To handle mathematical operations between two objects.

15. What is the output of the following code snippet?

```
def outer_function(x):
    def inner_function(y):
        return x + y
    return inner_function
closure = outer_function(10)
print(closure(5))
```

a) 10
b) 5
c) 15
d) TypeError

16. How can you implement a singleton pattern in Python?

a) By subclassing the Singleton base class provided in Python's standard library.
b) By making the constructor __init__() private.
c) By overriding the __new__() method to return an existing instance if one already exists.
d) By using a class variable and a static method to check for an existing instance.

17. What is a metaclass in Python?

a) A class that is used to define the behavior of other classes.
b) A base class for all other classes.
c) A class that is used to create instances of a class.
d) A special type of decorator for classes.

18. What is the purpose of the collections.namedtuple class?

a) To create an immutable list with named elements.
b) To create a dictionary with a fixed set of keys.
c) To create tuple subclasses with named fields, which makes the code more readable.
d) To create a list where elements can be accessed by a key.

19. What is the effect of using __name__ == '__main__' in a Python script?

a) It executes the code block only if the script is imported as a module.
b) It defines the main function of the script.
c) It ensures the code block is executed only when the script is run directly.
d) It is used to mark the end of the script.

20. What is the purpose of the __slots__ attribute in a class definition?

a) To prevent the creation of new instances of a class.
b) To pre-allocate memory for a fixed set of attributes, reducing memory consumption.
c) To make all attributes of a class private.
d) To make a class a singleton.